Better Magnets Using Electricity

An Electromagnet can be made by winding a piece of insulated wire around a piece of steel bar and connecting it to a power source (battery or power pack). Your task is to test several different designs of electromagnets to see which works best.

Test 1.

Does an electro-magnet work best using a small battery or a large battery?

Make your electro-magnet by winding the wire around the bolt and see how many paperclips it can pick up using the 'D' battery (larger one) and then the 'AA' battery (smaller one).

Number of paperclips picked up by the large battery: _____

Number of paperclips picked up by the small battery: _____

Which one worked best? _____ How much better was it? _____

Test 2.

Does an electromagnet work better with a few loops of wire or lots of turns of wire?

Make your electro-magnet (use the best battery from the previous test) and see how many paperclips it can pick up with 1 metre of wire wrapped around it (only use ½ the wire in loops) and then 2 meters of wire (wrap the whole wire around the bar).

Number of paperclips picked up magnet with 1 meter of loops: _____

Number of paperclips picked up magnet with 2 meter of loops: _____

Which one worked best? _____ How much better was it? _____

Test 3

Use the best number of loops and the best battery from the previous tests and make an electro-magnet using a plastic pen instead of an iron bar or bolt.

Did this type of electromagnet work better than a normal one with an iron centre?

Electromagnet Questions

1. Using your results from above, what features would a really powerful electromagnet have?

2. Give 2 reasons why are electromagnets more useful than normal magnets?

3. Try to think of 3 places that electro-magnets are used in everyday life.